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Magnetic spectral signatures in the Earth's magnetosheath and plasma depletion layer

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Abstract. Correlations between plasma properties and magnetic fluctuations in the subsolar magnetosheath downstream of a quasi-perpendicular shock have been found and indicate that mirror and ion cyclotronlike fluctuations correlate with the magnetosheath proper and plasma depletion layer, respectively (*Anderson and Fuselier*, 1993). We explore the entire range of magnetic spectral signatures observed from the AMPTE/CCE spacecraft in the magnetosheath downstream of a quasi-perpendicular shock. The magnetic spectral signatures typically progress from predominantly compressional fluctuations, $\delta B_{\parallel}/\delta B_{\perp} \approx 3$, with $F/F_p < 0.2$ (F and F_p are the wave frequency and proton gyrofrequency, respectively) to predominantly transverse fluctuations, $\delta B_{\parallel}/\delta B_{\perp} \approx 0.3$, extending up to F_p. The compressional fluctuations are characterized by anticorrelation between the field magnitude and electron density, n_e, and by a small compressibility, $C_e \equiv (\delta n_e/n_e)^2(B/\delta B_{\parallel})^2 \approx 0.13$, indicative of mirror waves. The spectral characteristics of the transverse fluctuations are in agreement with predictions of linear Vlasov theory for the H^+ and He^{2+} cyclotron modes. The power spectra and local plasma parameters are found to vary in concert: mirror waves occur for $\beta_{\parallel p}$ ($\beta_{\parallel p} \equiv 2\mu_0 n_p k T_{\parallel p}/B^2$) ≈ 2, $A_p \equiv T_{\perp p}/T_{\parallel p} - 1 \approx 0.4$, whereas cyclotron waves occur for $\beta_{\parallel p} \approx 0.2$ and $A_p \approx 2$. The transition from mirror to cyclotron modes is predicted by linear theory. The spectral characteristics overlap for intermediate plasma parameters. The plasma observations are described by $A_p = 0.85\beta_{\parallel p}^{-0.48}$ with a log regression coefficient of -0.74. This inverse $A_p - \beta_{\parallel p}$ correlation corresponds closely to the isocontours of maximum ion anisotropy instability growth, $\gamma_m/\Omega_p = 0.01$, for the mirror and cyclotron modes. The agreement of observed properties and predictions of local theory suggests that the spectral signatures reflect the local plasma environment and that the anisotropy instabilities regulate A_p. We suggest that the spectral characteristics may provide a useful basis for ordering observations in the magnetosheath and that the $A_p - \beta_{\parallel p}$ inverse correlation may be used as a beta-dependent upper limit on the proton anisotropy to represent kinetic effects.

1. Introduction

Although magnetosheath flow may be approximately described by gas dynamic models, the importance of magnetic forces is widely recognized, particularly near the magnetopause in the subsolar region [e.g., *Song et al.*, 1990b; *Wu*, 1992]. It has been theorized for some time that the subsolar magnetosheath should suffer a depletion of plasma density and an increase in magnetic field strength within about $0.5R_E$ of the magnetopause [*Midgley and Davis*, 1963; *Lees*, 1964; *Zwan and Wolf*, 1976; *Wu*, 1992]. The phenomenon is referred to as plasma depletion, and the region of reduced density and increased field strength is called the plasma depletion layer. Observational evidence has been reported to support this expectation [*Paschmann et al.*, 1978; *Crooker et al.*, 1979] and has indicated that the plasma depletion layer preferentially forms when the upstream shock is quasi-perpendicular and when the magnetosheath field is northward [*Anderson and Fuselier*, 1993] (hereinafter referred to as AF).

ULF fluctuations in the magnetosheath magnetic field indicative of both ion cyclotron and mirror instabilities have been found. *Fairfield* [1976] first noted that the magnetosheath near the magnetopause exhibited fluctuations near 1 Hz, below the local proton cyclotron frequency with a left-hand transverse nature suggestive of ion cyclotron waves. *Tsurutani et al.* [1982] have observed large-amplitude compressional fluctuations, $\delta B/B \approx 0.5$, in the magnetosheath that were attributed to the mirror instability. Subsequent authors also noted magnetic fluctuations characteristic of the magnetosheath near the magnetopause [*Song et al.*, 1990a, 1993; *Anderson et al.*, 1991; AF].

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Anderson et al. [1991] and AF demonstrated a close association between the wave magnetic fluctuations and plasma properties. They found that cyclotron waves occurred for reduced plasma density and increased magnetic field strength relative to values during mirror waves. Their observations also suggested that magnetosheath ion anisotropies may be ordered by the ion anisotropy mode that is most unstable.

Denton et al. [1993] compared the magnetic field spectral observations for one ion cyclotron wave event with theoretical predictions using the observed plasma properties and showed that the observed frequency and polarization structure of the magnetic fluctuations agree with theoretical expectations. The solar wind He^{2+} was found to play a significant role in determining the spectral structure. *Gary et al.* [1993a] used measured plasma data to show that the observed transition from mirror to cyclotron waves occurs where the growth rates are comparable and that the mode with the larger growth rate is observed to be dominant.

We extend the observational work of AF to consider the full range of spectral features observed downstream of the quasi-perpendicular shock. Since we use the Active Magnetospheric Particle Tracer Explorers/Charge Composition Explorer (AMPTE/CCE) data set, we necessarily restrict attention to the subsolar magnetosheath for which the magnetosphere is compressed. AF was primarily concerned with periods that were dominated by either compressional fluctuations or left-hand polarized transverse waves above $0.5F_p$. In general, a continuum of spectral signatures is observed between these extremes. Characterizing the variety of spectral signatures and corresponding plasma conditions is the topic of this paper.

Section 2 describes the instrumentation and preliminary data analysis. Section 3 presents an example magnetosheath pass that exhibits the range of spectral features to be examined. We define five spectral categories on the basis of the observed morphology. Section 4 presents the results of statistical analysis of 102 magnetosheath intervals. The results are discussed in terms of the ion cyclotron and mirror instabilities in section 5.

2. Data Analysis

The AMPTE/CCE spacecraft was in an equatorial orbit with magnetic latitude in the range ±16°, apogee at $8.8R_E$, perigee at 1000 km, and a period of 15.6 hours. During normal magnetospheric conditions the spacecraft remained within the magnetosphere throughout its orbit, but magnetospheric compressions occasionally pushed the magnetopause inward within the CCE apogee.

Magnetic field data are provided by the Magnetic Field Experiment [*Potemra et al.*, 1985], which included a vector flux gate magnetometer system with sensors mounted on a 2.3-m boom. The experiment had seven automatically switchable ranges from ±16 nT to ±65,536 nT full scale with 13-bit resolution ($\Delta B = 0.004$ to 16 nT) and obtained a vector sample every 0.124 s. Data were analyzed for ULF waves in a local field-aligned coordinate system.

Plasma data used here were from the Hot Plasma Composition Experiment (HPCE) [*Shelley et al.*, 1985], which consisted of an energetic ion mass spectrometer with an energy per charge range from the spacecraft potential to 17.1 keV/e and a set of separate electron spectrometers that measured the electron distribution in eight broad energy bands from 50 eV to 25 keV. The detectors were oriented to observe particles incident perpendicular to the spacecraft spin axis so that two-dimensional velocity coverage was provided by the spacecraft spin. The spin period was 6 s, and complete energy and species coverage for ions required approximately 2 min in the most favorable instrument mode. The electron spectrometers provided an energy spectrum every 0.155 s. The spin axis was oriented toward the sun. In the subsolar magnetosheath, the X component of the field is very small relative to the Y and Z components. As a result, the field was within 5° of the ion instrument field of view in 76 of the 102 intervals, whereas in four cases the minimum observed pitch angle was as large as 30°. In addition, plasma motion in the X direction should be much slower than the thermal speeds so that $T_\perp$ and $T_\parallel$ are reliably determined.

3. Example

Overview

Plate 1 shows plasma and magnetic field data from 0100 to 0700 UT on October 6, 1984, illustrating the range of magnetic spectra that are observed in the magnetosheath downstream of the quasi-perpendicular shock. Energetic solar wind ions are a signature of a quasi-parallel upstream shock [*Crooker et al.*, 1981; *Fuselier et al.*, 1991a; *Kucharek and Scholer*, 1991], so we use energetic, >10 keV, He^{2+} to infer whether the upstream shock is quasi-parallel or quasi-perpendicular. The left hand panels show spin-averaged electron number fluxes from the 67-eV and 8.9-keV channels and proton density n_p together with the magnetic field in GSM coordinates. Electron and vector field data are shown with 6-s resolution. Proton moments were determined with 2-min resolution. Magnetic fluctuation and proton distribution data are shown in the right-hand panels, which show from top to bottom: ellipticity of the transverse fluctuations from 0.03 to 4.0 Hz; transverse power; parallel power; proton temperature anisotropy, A_p; and proton plasma beta, $\beta_p \equiv 2\mu_0 n_p k T_p / B^2$, $T_p \equiv (2T_{\perp p} + T_{\parallel p})/3$. The black lines indicate the 30-s averaged local He^{2+} and H^+ gyrofrequencies, $0.5F_p$ in the second panel and F_p in the third panel.

The spacecraft was in the magnetosheath from about 0215 to 0640 UT as indicated by the low fluxes of 8.5-keV electrons and high fluxes of 67-eV electrons. (Low energy, 50–100 eV, and higher energy, 5–10 keV electrons are characteristic of the magnetosheath and magnetosphere, respectively.) From 0147 to 0215 UT, the spacecraft stayed near the magnetopause, perhaps even in the diffusion region. This is suggested by the variable field direction, high plasma density, and extraordinarily high β_p. Similar fluctuations are commonly observed in association with magnetopause crossings with high magnetic shear and have been associated with reconnection processes [*Song et al.*, 1993]. We are interested in the magnetosheath per se, so we turn our attention to the remainder of the magnetosheath interval.

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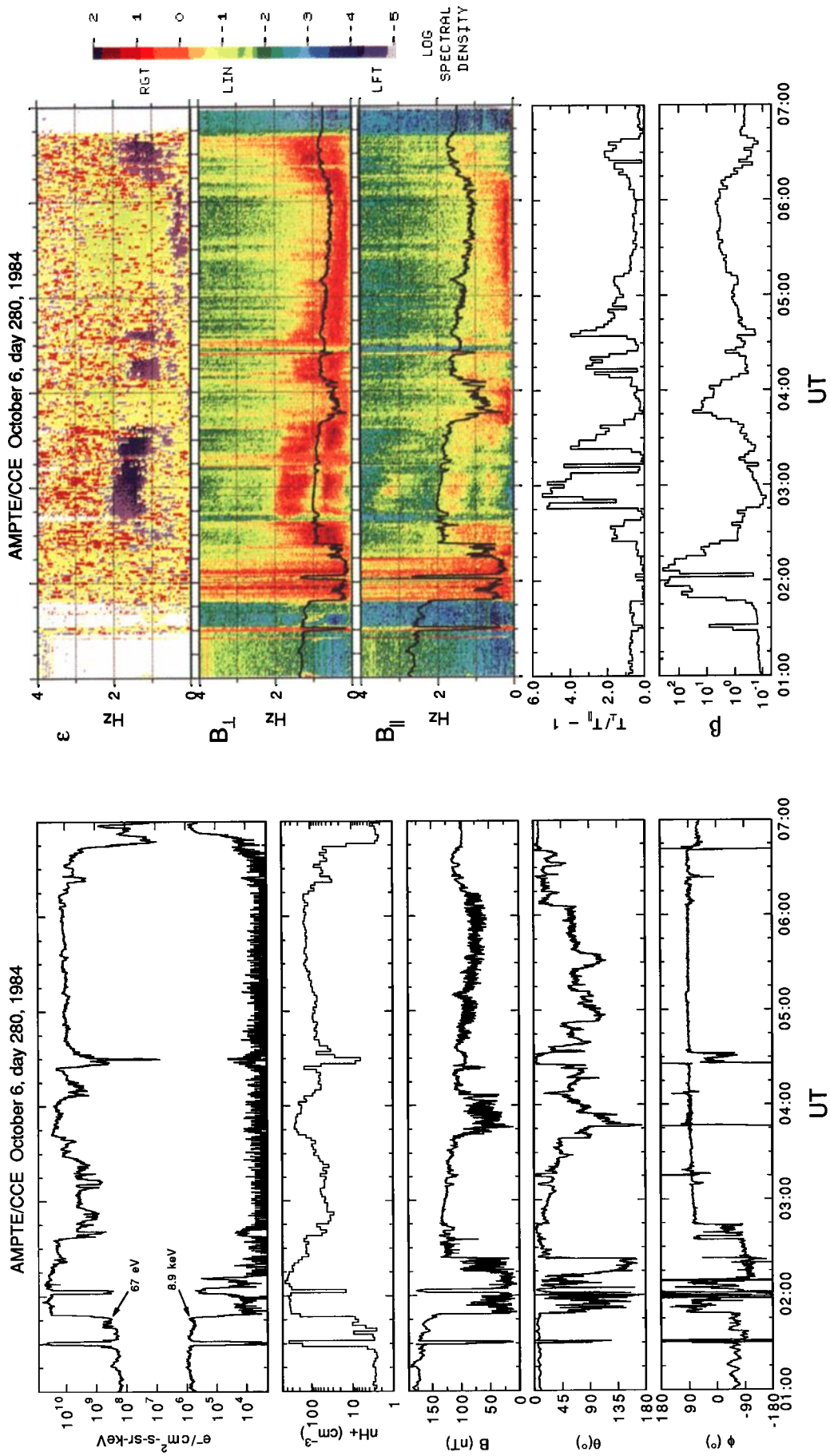

Plate 1. Particle and magnetic field data for the magnetosheath interval of October 18, 1984. Left panels from top to bottom: spin-averaged electron fluxes for the 67-eV and 8.9-keV channels, proton density, and 6-s magnetic field data in GSM coordinates: magnitude, polar (0° = north, 180° = south) and azimuth (0° = sunward, 90° = eastward) angles. Right panels from top to bottom: dynamic spectrogram of ellipticity in the plane perpendicular to **B**, transverse spectral power, parallel spectral power; $A_p \equiv T_{\perp p}/T_{\parallel p} - 1$; β_p. Ellipticity is displayed in three ranges: $\varepsilon \geq 0.2$ right-handed (red); $\varepsilon \leq -0.2$ left-handed (purple); and $|\varepsilon| < 0.2$ linear (yellow) and is shown only for those frequencies that exceed a certain power threshold appropriate to the digitization level of the instrument range. The black lines indicate the 30-s averaged local He^{2+} and H^+ gyrofrequencies, $0.5F_p$ in the second panel and F_p in the third panel.

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The variety of spectral signatures observed can be organized according to the relative intensity of the compressional signals and the spectral structure of the transverse power. Prominent compressional signals occur from 0215 to 0220 UT, from 0340 to 0400 UT, and from 0510 to 0600 UT. Their frequency is generally low, below $0.2F_p$, and they can reach rather large amplitudes, $\delta B_{\parallel}/B \approx 0.3$, as shown in the B trace. Predominantly transverse signals also occur: 0220–0240, 0245–0310, 0320–0340, 0400–0440, and 0615–0635 UT.

The transverse signals have a variety of frequency ranges and polarizations, and can occur simultaneously with compressional fluctuations. Transverse signals below $0.5F_p$ are often linearly polarized, whereas above $0.5F_p$ they are usually left-hand polarized, as during 0245–0300 UT. Exceptions are found, however, e.g., 0220–0240 UT (linear above $0.5F_p$) and 0530–0600 (left-handed below $0.5F_p$). The transverse signals are limited to frequencies below F_p. Transverse signals below $0.5F_p$ are always present when there are waves above $0.5F_p$ but not vice versa; compare 0400–0420 with 0510–0530 UT. Spectral slots in the transverse power near $0.5F_p$ appear occasionally and are evident from 0240 to 0340 UT. Strong compressional signals and transverse waves with $F > 0.5F_p$ seem to be mutually exclusive; note the transitions occurring at 0340 UT and 0400 UT. Transverse waves with $F < 0.5F_p$ do occur with compressional fluctuations, however, e.g., from 0510 to 0600 UT.

Progression of Spectral Types

The variety of spectral types appearing in this example can be viewed as a progression from compressional to transverse waves. In addition, the spectral types correspond to different local plasma conditions: high β_p–low A_p prevails for the strongly compressional fluctuations, whereas low β_p–high A_p characterizes the strongest transverse waves. The general inverse correlation of β_p and A_p is apparent in Plate 1 as is the correspondence of spectral type to β_p and A_p.

To provide a framework for quantitative analysis we define five spectral categories. The first category, mirror (MIR), represents instances when compressional perturbations dominate the spectrum while there is no discernible spectral peak in the transverse power. Note that the absence of a transverse spectral peak does not mean that the transverse power is negligible. We adopt the term mirror in anticipation of the discussion of wave modes presented below. The second category consists of intervals with a prominent compressional peak together with a peak at a different frequency in the transverse spectrum with $F < 0.5F_p$ and is called MRL for mirror with low-frequency transverse waves. Events with compressional and transverse peaks at the same frequency were considered to be compressional if the compressional and transverse powers were comparable. This situation arose in one event. The remaining categories consist of intervals with no discernable peak in compressional power but with three different transverse spectral structures: low frequency (LOW), transverse waves with $F < 0.5F_p$; continuous (CON), transverse waves with continuous power spectra extending above and below $0.5F_p$; and bifurcated (BIF), transverse waves with two broad local maxima in spectral power.

Figure 1 shows power spectra for five intervals of Plate 1, one spectrum for each category. The time intervals used to evaluate the power spectra are indicated in the panels, and the plasma parameters are given in Table 1. The minimum frequencies of these spectra, given by the reciprocal of the time spans, are considerably lower than shown in the spectrogram of Plate 1. The spectral density is displayed in left-hand, P_L, right-hand, P_R, and parallel, $P_{\parallel}$, components versus frequency normalized to F_p. To indicate the frequency ranges for which the linear Vlasov theory predicts positive ion cyclotron wave growth, the frequencies for which the ion cyclotron modes have $\gamma/\Omega_p > 10^{-4}$ are shown by the horizontal bars. The calculations were performed using the WHAMP dispersion solver [see *Denton et al.*, 1993]. Nonlinear and propagation effects may change the spectral structure somewhat, but the ranges of positive growth generally correspond to the frequencies of transverse spectral enhancements.

In the mirror spectrum a peak in $P_{\parallel}$ extends to the lowest frequencies, and P_R and P_L display no prominent peaks above $P_{\parallel}$. The compressional peak intensity is greater for the mirror case than any of the others. Linear theory predicts positive ion cyclotron wave growth, suggesting that the absence of a transverse spectral peak may not necessarily imply the absence of cyclotron waves.

The examples of LOW and MRL spectra illustrate transverse spectral enhancements below $0.5F_p$. In the MRL case, transverse enhancements are accompanied by a $P_{\parallel}$ peak that maximizes at a lower frequency than the transverse spectral peak. In the LOW case, $P_{\parallel}$ does not have a clear spectral peak. The predicted ranges of positive ion cyclotron mode growth correspond to the frequency ranges of transverse spectral enhancements.

The continuous case has a broad transverse spectral enhancement extending above $0.5F_p$. The portion of this peak above about $0.5F_p$ is predominantly left-hand polarized, $P_L > P_R$. The parallel power is relatively featureless as in the LOW example. Linear theory predicts two closely spaced bands of ion cyclotron wave growth. The transverse spectral peaks of the BIF, LOW, and MRL examples extend over broader frequency ranges than the $\gamma/\Omega_p > 10^{-4}$ criterion, so it is not surprising that the observations exhibit a continuous spectral peak in this case.

Finally, the example bifurcated case has a depression in wave power extending from about 0.35 to $0.55F_p$. P_L decreases rapidly with frequency from about 0.9 to $1.0F_p$. The left-handed polarization for $F > 0.5F_p$ and the mixed polarization of lower frequencies are evident. Linear theory predicts two frequency ranges of ion cyclotron wave growth coincident with the frequencies of the spectral peaks. In this case the bands of positive cyclotron wave growth with $\gamma/\Omega_p > 10^{-4}$ are separated by more than twice their width.

The categories defined here correspond to a continuum of spectral signatures ordered as MIR–MRL–LOW–CON–BIF. Although the distinctions between adjacent categories, LOW and MRL, for example, can be a matter of judgment, there is a clear distinction between non-neighboring categories, for example, MIR and LOW, or LOW and BIF.

These categories are somewhat different than those used by AF. The events of AF with left-hand polarization and $F > 0.5F_p$ are the BIF and CON categories, except that a

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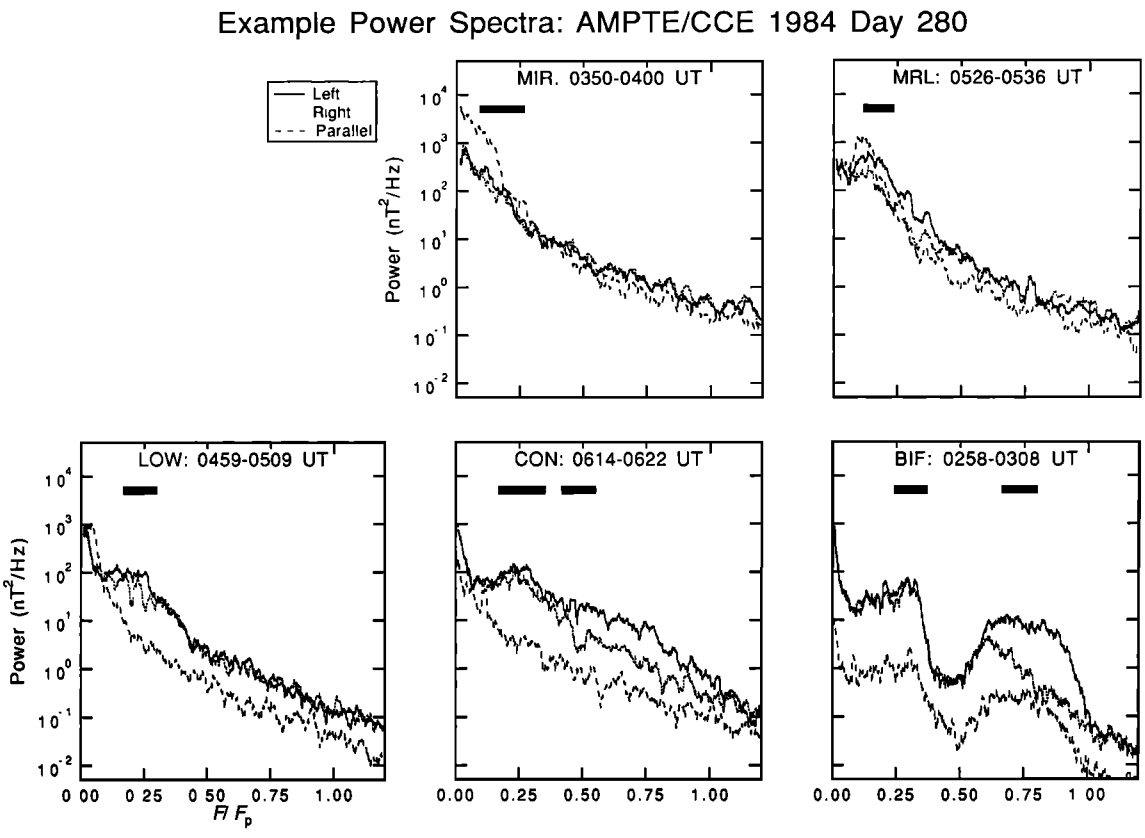

Figure 1. Example power spectra of magnetic field data. The five panels show power spectra corresponding to the five spectral categories defined for the statistical analysis. Left- and right-hand power in the plane perpendicular to the local field direction are shown together with the spectral density parallel to the field. Frequency is normalized to the local proton gyrofrequency. Horizontal bands represent the theoretical range of ion cyclotron instability as indicated by $\gamma \geq 10^{-4}\ \Omega_p$.

handful of continuous events did not have left-hand polarization above $0.5F_p$ and hence did not appear in AF (e.g., the 0220–0240 interval of Plate 1). Transverse waves of AF are the LOW events. These events were not treated as firmly identified ion cyclotron waves in AF because of the absence of left-hand polarization. The mirror waves of AF are the MIR events. The MRL category of this paper, mirror waves with low-frequency transverse waves, was excluded from analysis in AF.

Table 1. Average Plasma Parameters for Example Intervals

	Spectral Type				
Parameter	MIR	MRL	LOW	CON	BIF
UT start	0350	0524	0458	0614	0258
UT stop	0400	0538	0510	0622	0309
n_p (cm^{-3})	198	125	74.4	84.1	50.7
$T_{\perp p}$ (keV)	0.268	0.291	0.247	0.207	0.168
$T_{\parallel p}$ (keV)	0.200	0.189	0.103	0.093	0.041
A_p	0.36	0.54	1.41	1.27	3.98
β_p	6.8	2.37	0.787	0.67	0.165
B (nT)	58.4	76.2	89.9	98	128.9

The plasma parameters of Table 1 exhibit a progression from MIR to BIF categories. The bifurcated interval has the largest A_p and lowest β_p, whereas the mirror interval occupies the opposite extreme. The other categories are ordered in both A_p and β_p, from BIF to MIR (with the exception of A_p for LOW and CON). AF also noted that the upper frequency left-handed waves had systematically higher A_p and lower β_p than mirror wave periods. The new feature here is that the remainder of spectral features falls along the continuum between the extremes.

Wave Modes

Anderson et al. [1991] and AF suggested that the transverse waves having $F > 0.5F_p$ and left-hand polarization were ion cyclotron waves and the low-frequency compressional signals were mirror waves. *Gary et al.* [1993a] used the observed plasma data for the events of AF to show that the linear theory predicts that the mirror instability has the larger growth rate for events dominated by compressional power, whereas the ion cyclotron growth rate was greater for intervals dominated by transverse signals.

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The frequency structure of the transverse fluctuations is predicted by linear Vlasov theory of the ion cyclotron modes. Since He^{2+} in the magnetosheath is hotter than H^+ [*Fuselier et al.*, 1991b; *Gary et al.*, 1993c], we expect that He^{2+} may be generating cyclotron waves below $0.5F_p$. Hence the bifurcation feature is probably not analogous to the spectral gap observed in the magnetosphere, which is attributed to absorption by cold He^+ [e.g., *Fraser*, 1985]. The fact we observe continuous spectra also indicates that the slot is not due to absorption [*Denton et al.*, 1993; *Gary et al.*, 1993b]. The analysis of *Denton et al.* [1993] shows that He^{2+} is largely responsible for the lower spectral peak and that protons drive waves with frequencies above $0.5F_p$. The polarization structure of the transverse fluctuations, linear below $0.5F_p$ but left-handed above, can also be accounted for by the ion cyclotron modes [*Denton et al.*, 1993].

Although the growth rate analysis of *Gary et al.* [1993a] shows that the compressional fluctuations are consistent with the mirror mode, analysis of the density fluctuations allows a more stringent test of the wave mode. Fast, slow, and mirror modes can all be predominantly compressional. The density perturbations δn_e associated with these modes are different, however. For the fast mode, δB and δn_e are in phase, whereas they are in antiphase for the slow and mirror modes. The mirror and slow modes in turn may be distinguished using $C_e \equiv (\delta n_e/n_e)^2(B/\delta B_\parallel)^2$, since $C_e < 1$ for the mirror mode [*Gary*, 1992] but $C_e > 1$ for the slow mode at oblique propagation under magnetosheath conditions [*Gary and Winske*, 1992]. That is, magnetic field perturbations predominate in the mirror mode, but the density perturbations predominate in the slow mode.

To characterize the density fluctuations, we use the electron density sampled by HPCE. Since HPCE measures electrons above 50 eV, it is important to distinguish between the total electron density, $n_e = n_p + 2n_\alpha$, and the density sampled by HPCE, which we denote as N_e. In the magnetosheath the electron temperature T_e is typically 30–70 eV [*Feldman*, 1985], so HPCE misses a significant part of the distribution. Nonetheless, N_e is intimately related to n_e as well as the electron temperature, T_e. In particular, T_e and N_e are positively correlated because an increase in T_e brings more of the distribution into the instrument's energy range. Furthermore, in response to compressional fluctuations, T_e and n_e both change. If the electron response is adiabatic, $T_e/n_e^{(\gamma-1)}$ will be constant so that δn_e and δT_e have the same sign. Writing

$$\delta N_e = \frac{\partial N_e}{\partial n_e}\delta n_e + \frac{\partial N_e}{\partial T_e}\delta T_e \qquad (1)$$

we see that $\delta N_e/N_e$ is an overestimate of $\delta n_e/n_e$. Observationally, one can test that T_e and N_e are positively correlated by evaluating the average energy in the HPCE electron energy range, $\langle E \rangle$. For the MIR example, δN_e and

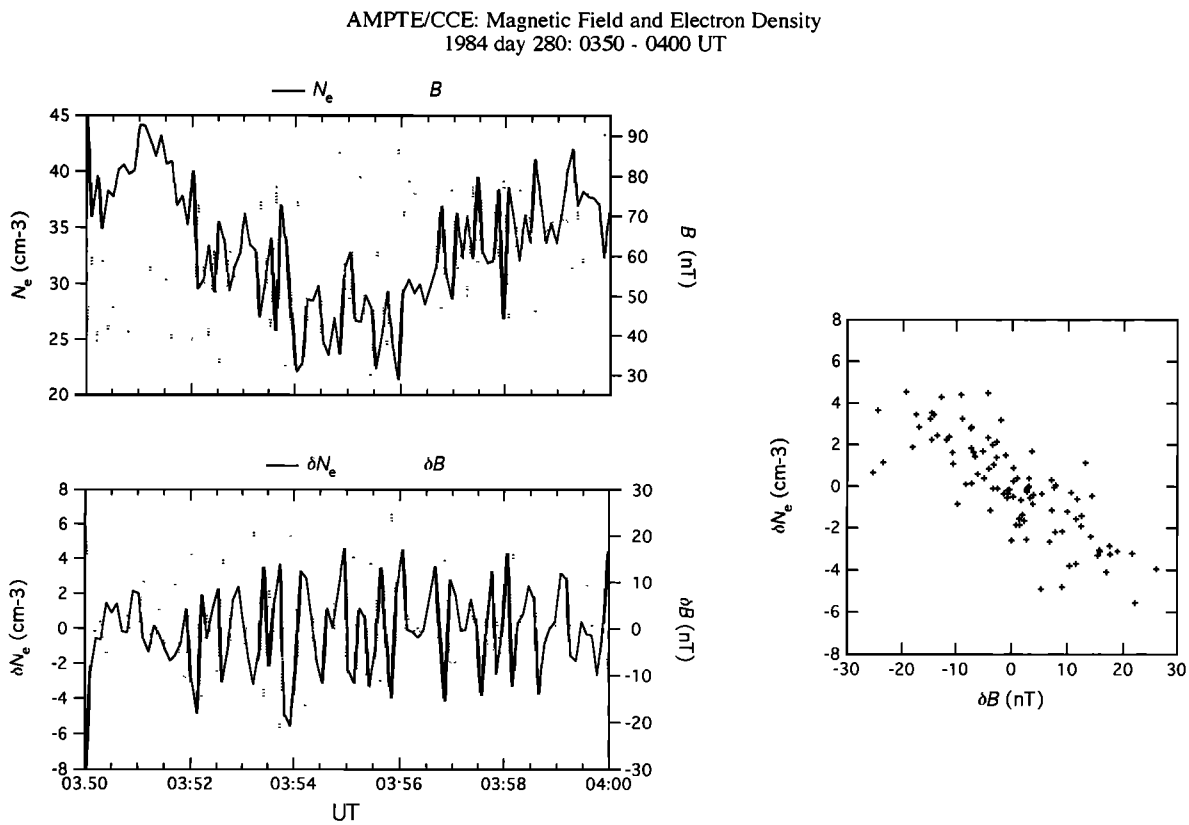

Figure 2. Magnetic field and electron density fluctuations for the 0350–0400 UT interval of 1984 day 280. Average magnetic field strengths were evaluated over the identical intervals used to spin average the electron data. Anticorrelation of field and density fluctuations is particularly apparent in the comparison of first differences, δN_e and δB. The linear regression coefficient, R_{NB}, between δN_e and δB is –0.80.

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$\delta\langle E\rangle$ are positively correlated with a regression coefficient of +0.70.

For the correlation analysis the field magnitude was averaged over the same accumulation intervals as the electron data. These data are shown in Figure 2 for the 0350 to 0400 UT interval. Both the time series and the first differences, that is, $\delta N_e = N_{ej+1} - N_{ej}$, of N_e and B are shown. The linear regression coefficient, R_{NB}, between δN_e and δB is –0.80. The anticorrelation indicates that the signal is not due to fast mode waves.

To test whether the signals are mirror or slow waves, we examine the compressibility C_e. To evaluate C_e, we need to quantify the relationship between $\delta n_e/n_e$ and $\delta N_e/N_e$. As shown in Appendix A, one can evaluate the sensitivity of the measured distribution to increases in temperature assuming only that the distribution is of the form $f(E/T_e)$ and obtain an expression for $(\delta N_e/N_e)/(\delta n_e/n_e)$ in terms of the measured electron distribution. For the data plotted in Figure 2 the average values for $\delta N_e/N_e$ and $\delta B/B$ were 0.11 and 0.27, respectively. Taking $\gamma = 5/3$ we find $(\delta N_e/N_e)/(\delta n_e/n_e) = 1.76$, which gives $C_e = 0.054$. The linear theory of *Gary and Winske* [1992] and *Gary* [1992], which should apply for the modest wave amplitudes of this case, predicts that $C_e < 1$ corresponds to the mirror instability, not the slow mode.

4. Statistical Analysis

Data Selection and Analysis

There were 14 magnetosheath passes with complete plasma and magnetic field data (cf. Table 1 of AF). In this study we are concerned with magnetosheath conditions downstream of a quasi-perpendicular shock, so we excluded intervals that had enhanced energetic He^{2+} fluxes [*Crooker et al.*, 1981; *Fuselier et al.*, 1991a; *Kucharek and Scholer*, 1991].

Dynamic spectra plotted with 2 hours of data per page were used to identify all periods longer than 5 min that exhibited steady spectral structure. We selected no more than one interval from each quarter hour. If the spectral features varied, multiple intervals were allowed in 15 min since these represented independent measurements of each spectral type. The interval durations also varied depending on whether the HPCE ion instrument was in a rapid or slow mode. The relative abundance of spectral types in the CCE data is not reflected in this analysis.

Line plots of power spectra, as shown in Figure 1, were then prepared for each interval, using up to a maximum of 10 min of data. These plots were examined to make the final category identifications. The middle 10 min of data was used for intervals longer than 10 min. No interval was shorter than 4 min, so the spectral resolution was no coarser than 4 mHz, 1/(4 min).

Plasma data were prepared by evaluating moments of H^+ and He^{2+} distributions. The following were determined for both ion species: n_j, $T_{\perp j}$, $T_{\parallel j}$, $A_j \equiv T_{\perp j}/T_{\parallel j} - 1$, $\beta_{\parallel j} \equiv 2\mu_0 n_j k T_{\parallel j}/B^2$, and $\beta_{\perp j} \equiv 2\mu_0 n_j k T_{\perp j}/B^2$. (Our recent papers employ various definitions of $\beta_\parallel$. We bring the definitions together in Appendix B). These quantities were computed at the highest time resolution available (2 min in most cases) and then averaged over each interval.

There were 102 intervals totaling 34 hours of observations, distributed among the categories as shown in Table 2. It was found that systematic differences between the He^{2+} and H^+ parameters were much less pronounced than the variations between categories. We therefore present only the H^+ data here.

Power spectra were analyzed in two ways. To determine whether the spectral signatures of Figure 1 reflect the statistical properties, we prepared composite power spectra (see below). We also evaluated the compressional and transverse fluctuation amplitudes, $\delta B_\parallel$ and $\delta B_\perp$, respectively, where

$$\delta B^2_{\parallel,\perp} = \int_{F_0}^{F_1} P_{\parallel,\perp}\, dF \qquad (2)$$

P is the spectral power in nT^2/Hz, and $P_\perp \equiv P_L + P_R$. The frequency limits F_0 and F_1 in (2) were chosen after examination of the composite power spectra (see below): 0.0–0.2F_p for mirror, and 0.1–1.0 F_p for ion cyclotron waves.

Compressional Mode Analysis

Electron density and magnetic field magnitude fluctuations were analyzed for the MIR and MRL intervals. Figure 3 shows the linear regression coefficient between δN_e and δB, R_{NB}, versus $\delta N_e/N_e$, together with a histogram distribution of R_{NB}. The mean of R_{NB} was –0.70 and the me-

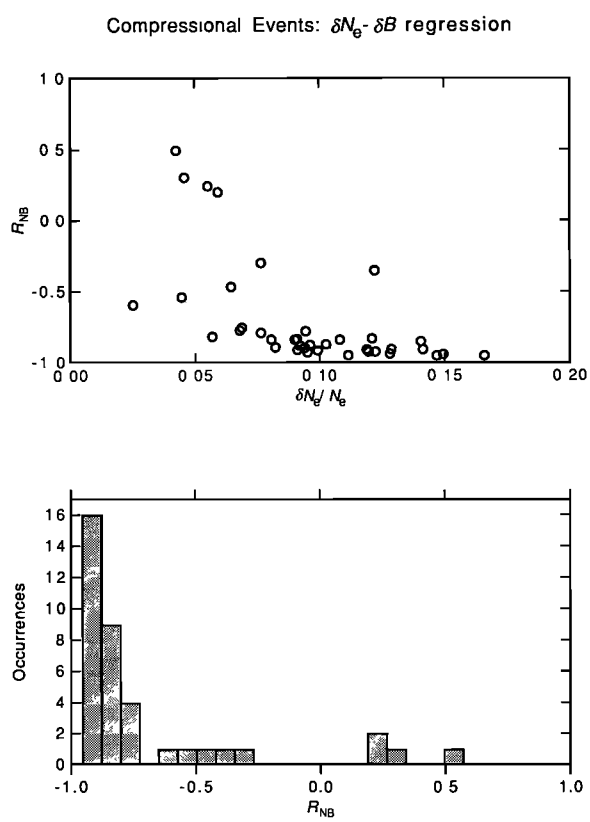

Figure 3. Linear regression coefficients between δN_e and δB versus the average $\delta N_e/N_e$ for the MIR and MRL events. The predominance of anticorrelation between δN_e and δB is apparent. The points with less pronounced correlation occur when $\delta N_e/N_e$ is low and hence when the regression analysis is less reliable.

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dian was –0.84, indicating anti-correlation between δN_e and δB. Of the 38 events, 34 had $R_{NB} < 0$, and 29 had $R_{NB} < -0.7$. The scatterplot shows that positive R_{NB} occurred for smaller $\delta N_e/N_e$, and that the predominance of negative R_{NB} increased with $\delta N_e/N_e$. We conclude that the compressional fluctuations represented by the MIR and MRL events are not fast mode waves.

To test that density and temperature were positively correlated, regressions between δN_e and $\delta\langle E\rangle$ were done. The average regression coefficient between δN_e and $\delta\langle E\rangle$ was +0.81; the median was +0.93. In no case were δN_e and $\delta\langle E\rangle$ anticorrelated. For $T_e/n_e^{\gamma-1}$ constant, $\delta T_e/T_e = (\gamma - 1) \times \delta n_e/n_e$ should hold. The average of $(\delta\langle E\rangle/\langle E\rangle)/(\delta n_e/n_e)$ was 0.56, which is close to 2/3, indicating that the response is nearly adiabatic with $\gamma = 5/3$. We conclude that the data are consistent with adiabatic electron behavior.

The results of the compressibility analysis are presented in Figure 4. The average value of $(\delta N_e/N_e)/(\delta n_e/n_e)$ was 2.25. The line shows $\delta n_e/n_e = \delta B/B$; all but one of the 38 points fall below this line. The bottom panel shows the histogram distribution of C_e, which is heavily weighted toward $C_e < 1$. The average of $\log(C_e)$ was –0.87, $C_e = 0.13$; the median was -0.81, $C_e = 0.15$. The predominance of small C_e indicates that the events are consistent with the mirror mode but not the slow mode. The data imply this conclusion even without the correction factor for $(\delta N_e/N_e)/(\delta n_e/n_e)$. This result should not be taken to imply that slow modes cannot occur in the magnetosheath, since the MRL and MIR events represent those intervals for which the compressional perturbation produces the most prominent spectral peak. A general discussion of the nature of compressional fluctuations in the magnetosheath is beyond the scope of this paper.

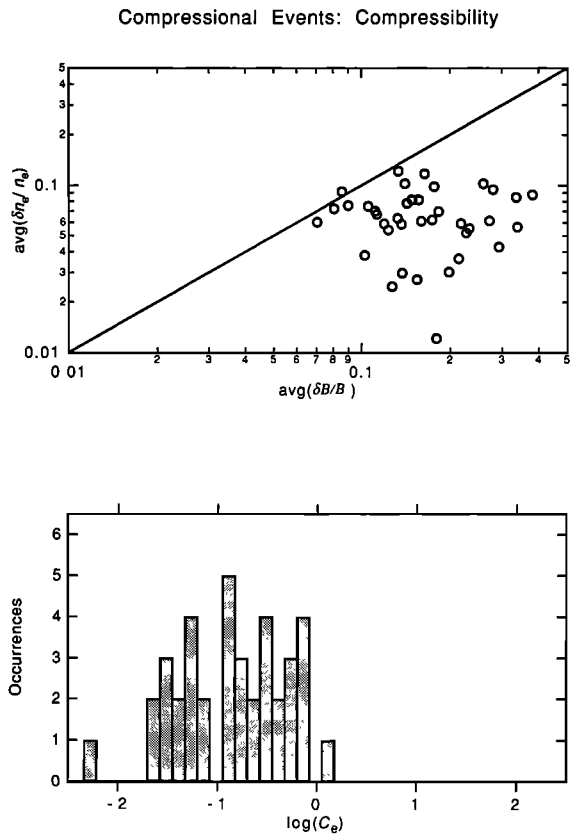

Figure 4. Histogram distribution of parallel compressibility, C_e, for the MRL and MIR intervals. The correction factor to obtain $\delta n_e/n_e$ from $\delta N_e/N_e$ has been applied to each interval using the electron temperature obtained using (A5). The parallel compressibility is predominantly less than unity, the average and median compressibilities are $\log(C_e) = -0.87$ ($C_e = 0.13$) and $\log(C_e) = -0.81$ ($C_e = 0.15$), respectively. The predominance of $C_e < 1$ indicates that the mirror mode predominates in the intervals selected for this analysis. The absence of any point for which $C_e >> 1$ indicates that no case presented unmistakable signatures of the slow mode.

Composite Power Spectra

Figure 5 shows the composite power spectra for each category. Because the background field varied between events in the same category and by nearly a factor of 2 between different categories, these spectra are not direct averages of spectral power density. Instead, we used the following procedure.

Frequency was divided into 50 bins from 0 to $1.0F_p$ in steps of $0.02F_p$. The lowest frequency sampled, F_{min}, ranged from 1.6 mHz (10-min intervals) to 4 mHz (4-min intervals). This corresponds to $F_{min}/F_p = 0.0023$ to 0.0053 for $B = 50$ nT, and $F_{min}/F_p = 0.0011$ to 0.0026 for $B = 100$ nT. Each of the 50 bins therefore contains between 3 and 20 spectral samples depending on the background field strength. The square amplitudes of left, right, and parallel power were evaluated over each bin using (2). The resulting square amplitudes were divided by 0.02 to yield a spectral density, $P^*(X)$, in nT^2 giving the mean square amplitude per unit proton gyrofrequency. That is,

$$P^*(X) = \frac{1}{0.02}\int_{(X-0.01)F_p}^{(X+0.01)F_p} P(F)dF \qquad (3)$$

We then averaged $\log(P^*)$ over each category to obtain the spectra of Figure 5. Averaging $\log(P^*)$ rather than P^* ensured that all events were given equal weight.

The composite spectra are similar to those of Figure 1, indicating that the spectral characteristics hold in general. One difference is that for BIF and CON events, the left-hand power is generally greater than right-hand power and becomes more dominant with increasing frequency. In any case the evolution of spectral character from MIR to BIF is the same as found in the example.

Average Quantities

Table 2 presents the average proton plasma moments and relative fluctuation amplitudes, $\delta B_{\parallel}/B$ and $\delta B_{\perp}/B$. The average A_p increases monotonically from MIR to BIF, whereas the average $\beta_{\parallel p}$ decreases monotonically. Both $\delta B_{\parallel}/B$ and $\delta B_{\perp}/B$ decrease from MIR to BIF, but the decrease in $\delta B_{\parallel}/B$ is more pronounced.

Anisotropy–Beta Correlation

Figure 6 shows A_p versus $\beta_{\parallel p}$ for all of the categories, plotted separately and together. The proton anisotropy was correlated slightly better with $\beta_{\parallel p}$ than with β_p. The inverse correlation between A_p and $\beta_{\parallel p}$ is apparent. The solid line shows the least-squares fit: $\log(A_p) = (-0.070 \pm 0.017) -$

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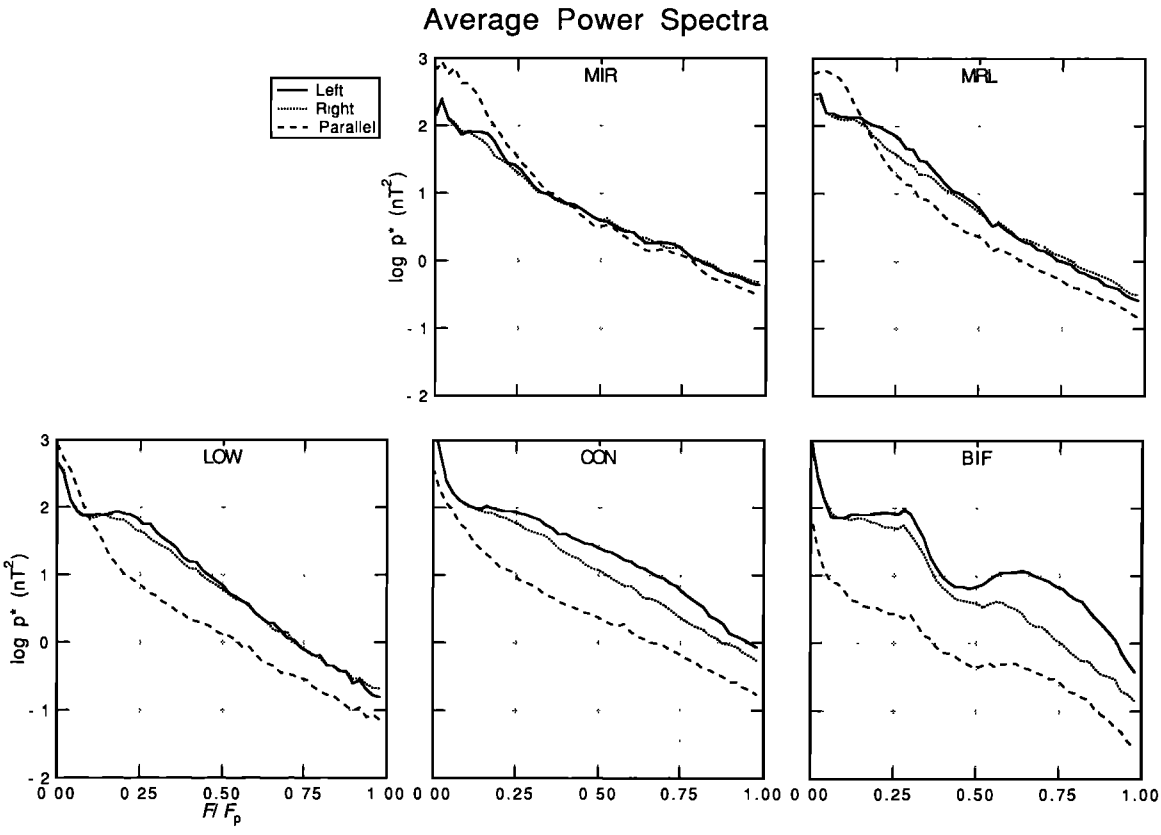

Figure 5. Average power spectra for the five categories used in statistical analysis. Left, right, and parallel spectral density are defined as in Figure 1. Frequency is plotted from 0 to 1 F_p. The spectra were averaged in normalized frequency rather than in real frequency to avoid smearing of spectral features. The spectral intensity has been normalized to yield the mean square amplitude per unit F_p. See text for additional details.

$(0.480 \pm 0.031)\log(\beta_{\|p})$, with a regression coefficient of –0.74. The data points were weighted by the standard errors in the fit. Points with low A_p tended to have large uncertainties in $\log(A_p)$, which is why the fit lies toward the upper edge of the data. This fit is a quantitative empirical relation that may be used to relate $T_{\perp p}$, $T_{\|p}$, and $\beta_{\|p}$ in models of magnetosheath flow as described below.

There is a mapping between spectral type and location along the $A_p - \beta_{\|p}$ curve. This indicates that the spectral features reflect the local plasma conditions, and hence that the observed evolution of spectral signatures results from changes in the plasma. In addition, it was shown in AF that the BIF and CON categories are indicative of the plasma depletion layer, whereas the MIR category corresponds to the magnetosheath proper. Since the opposite ends of the $A_p - \beta_{\|p}$ curve represent extremes in spatial location, the inverse correlation suggests that the plasma follows a well-defined path in $A_p - \beta_{\|p}$ space as it convects from the bow shock toward the magnetopause.

Figure 7 shows the data for all events together with isocontours of maximum growth rate from the Vlasov dispersion equation using the parameters of Table 1 of *Gary et al.* [1993a], $n_\alpha/n_e = 0.04$ and $T_{\perp\alpha}/T_{\|\alpha} = 1.25T_{\perp p}/T_{\|p}$. The dashed curve is the contour of maximum growth rate $\gamma_m = 0.01\Omega_p$ for the proton cyclotron anisotropy instability, and the dashed–dotted curve shows the same contour for the mirror instability. For most of the $\beta_{\|p}$ range considered here, the corresponding contour for the helium cyclotron anisotropy instability is very similar to that of the proton cyclotron mode.

The $A_p - \beta_{\|p}$ inverse correlation is closely related to the growth conditions for the ion anisotropy instabilities. The fact that $\gamma_m = 0.01\Omega_p$ lies closer to the data than the threshold curves presented in Figures 6 and 7 of *Gary et al.* [1993a] indicates that the observed $A_p - \beta_{\|p}$ relation corresponds to finite positive growth rate. Most importantly, the correspondence between observations and theory indicates that the fluctuating fields of the instabilities scatter the ions to yield the $A_p - \beta_{\|p}$ inverse correlation [*Gary and Winske*, 1993].

Amplitudes

In addition to wave spectral structure and the $A_p - \beta_{\|p}$ relation, the fluctuation amplitudes provide a useful point of comparison with theory. Linear theory does not predict wave amplitudes, but simulations do yield fluctuation amplitudes. Since frequency spectra are difficult to obtain from simulations, it is useful to calculate the total fluctuation amplitudes.

Figures 8 and 9 show the amplitudes δB and relative amplitudes $\delta B/B$ of the parallel and transverse fluctuations versus $\beta_{\|p}$. The frequency range appropriate to mirror fluctuations, 0 to 0.2 F_p (cf. Figure 5), was used in (2) to

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Table 2. Average Parameters for Each of the Five Spectral Categories

Parameter (Protons)	Spectral Type				
	MIR	MRL	LOW	CON	BIF
Number of intervals	13	25	19	29	16
Duration, min	17.2	25.7	17.9	19.4	17.8
β_p	7.4	3.36	1.84	1.65	0.45
(±)	1.3	0.37	0.19	0.27	0.07
$\beta_{\parallel p}$	6.15	2.52	1.23	1.11	0.22
(±)	1.29	0.30	0.15	0.18	0.05
Anisotropy	0.39	0.55	0.83	0.96	2.14
(±)	0.04	0.07	0.05	0.08	0.25
Density (cm^{-3})	148	133	128	80	76
(±)	17	17	22	13	9
B (nT)	55.7	75.4	89.2	89.1	114.1
(±)	3.5	2.5	6.7	4.5	3.6
$\log(\delta B_\parallel/B)$:	−0.54	−0.76	−0.99	−1.19	−1.75
(±)	0.02	0.02	0.03	0.05	0.08
$\log(\delta B_\perp/B)$:	−0.94	−1.11	−1.07	−1.0	−1.2
(±)	0.04	0.02	0.03	0.02	0.03

Duration is the length of time used to average the plasma parameters. Anisotropy is $T_\perp/T_\parallel - 1$. The fluctuation amplitudes are defined as described in the text using equation (3): $\delta B_\parallel$ is the rms amplitude of parallel fluctuations from 0 to $0.2F_p$; $\delta B_\perp$ is the rms amplitude of perpendicular fluctuations from 0.1 to $1.0F_p$. Uncertainties are standard errors in the mean values.

evaluate $\delta B_\parallel$. Solid circles show MIR and MRL events, whereas other categories are indicated by open circles.

Both $\delta B_\parallel$ and $\delta B_\parallel/B$ increase monotonically with $\beta_{\parallel p}$; the $\delta B_\parallel/B$ data are somewhat more tightly clustered together. The MIR and MRL events exhibit a different variation of $\delta B_\parallel/B$ than the other events. To illustrate this, we performed two fits of $\delta B_\parallel/B$ vs $\beta_{\parallel p}$, one including only the BIF, CON, and LOW categories and a second including only the MIR and MRL categories. The fit to nonmirror mode events gave $\delta B_\parallel/B \propto \beta_{\parallel p}^{0.33}$, whereas the fit using only the MIR and MRL events gave $\delta B_\parallel/B \propto \beta_{\parallel p}^{0.69}$. The difference indicates that different mechanisms determine $\delta B_\parallel/B$ in the two sets of events.

Corresponding plots of transverse amplitude versus $\beta_{\parallel p}$ are given in Figure 9. We chose the 0.1 to 1.0 F_p bandwidth since this represents the frequency range of the transverse waves (cf. Figure 5). Open circles were used for MIR events and filled circles for all others. Both $\delta B_\perp$ and $\delta B_\perp/B$ increase slightly with $\beta_{\parallel p}$, but the dependence is weaker than for $\delta B_\parallel$. We find $\delta B_\perp/B << 1$ consistent with previous observations of ion cyclotron fluctuations in the magnetosheath [*Sckopke et al.*, 1990]. The low amplitudes do not mean that cyclotron waves are unimportant for the particle dynamics since ion cyclotron instabilities have a dramatic effect on A_p even for low amplitudes [*Gary et al.*, 1993b].

Bifurcation

To study the spectral slot feature we first removed the background $1/F^2$ spectral slope by differencing the data (see *Hamming* [1983] or *Takahashi et al.* [1990] for a discussion of differencing). The location of the slot is defined

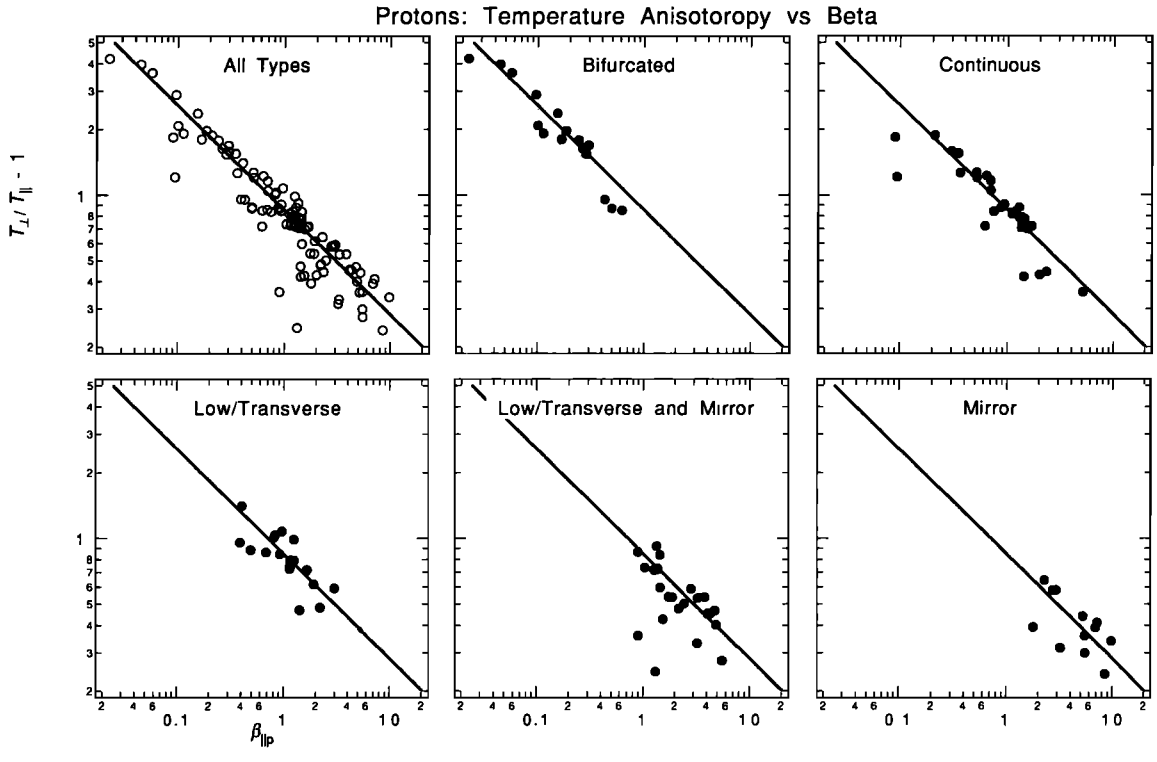

Figure 6. Plot of proton temperature anisotropy versus proton parallel beta for all events combined and each spectral category individually. The solid line shows the least-squares fit relation: $A_p = 0.85\beta_{\parallel p}^{-0.48}$. A progression of spectral type with location in A_p–$\beta_{\parallel p}$ space is apparent.

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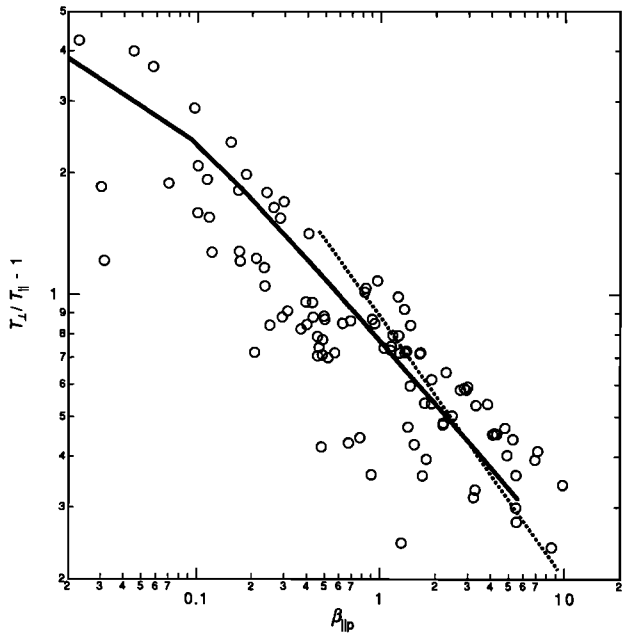

Figure 7. Plot of proton temperature A_p versus $\beta_{\parallel p}$ for all events together with isocontours of constant growth rate for the proton cyclotron and mirror instabilities. The dashed curve shows the contour of proton cyclotron growth rate of $0.01\Omega_p$, and the dotted curve shows the corresponding contour for the mirror mode assuming 4% He^{2+}. The data fall along a line essentially parallel to the contours of constant instability growth, indicating that the plasma characteristics are closely tied to the waves.

as the midpoint between the half-minimum frequencies. The depth is defined as the difference between the minimum log spectral density in the slot and the average of the maximum log spectral density at frequencies above and below the slot.

The bottom panel of Figure 10 shows the depth of the slot versus $\beta_{\parallel p}$. The bifurcation becomes more pronounced as $\beta_{\parallel p}$ decreases. This is consistent with the development from continuous to bifurcated spectra as $\beta_{\parallel p}$ decreases (cf. Figure 5). The top panel shows the distribution of the slot location with F/F_p. The data generally fall between 0.4 and 0.5 and are centered on 0.45. In analysis of the event of *Anderson et al.* [1991], *Denton et al.* [1993] found that the proton cyclotron mode generated waves only above $0.5F_p$, whereas the He^{2+} cyclotron mode generated waves below about $0.4F_p$, leaving a spectral slot that was centered at $0.45F_p$.

5. Discussion

Theoretical Interpretation

Linear Vlasov theory predicts that for the anisotropic, $T_\perp > T_\parallel$, H^+ and He^{2+} of the magnetosheath, three distinct growing modes may arise: mirror, proton cyclotron anisotropy, and helium cyclotron anisotropy instabilities [*Gary et al.*, 1993a]. The mirror instability has zero real frequency in the plasma center-of-mass frame, propagates with wave vector substantially oblique to the background magnetic field B, and has a fluctuating magnetic field predominantly in the direction parallel to B [*Gary*, 1992, and references therein]. The two cyclotron instabilities have nonzero real frequencies bounded above by their respective cyclotron frequencies, maximum growth rates at wave vectors parallel to B, and fluctuating magnetic fields predominantly perpendicular to B [*Gary et al*, 1993a; *Denton et al.*, 1993]. That the cyclotron modes are responsible for the transverse fluctuations is indicted by the agreement between the predicted and observed transverse spectral structures. With regard to the compressional fluctuations, detailed correlation analysis of the density and field perturbations implicates the mirror mode.

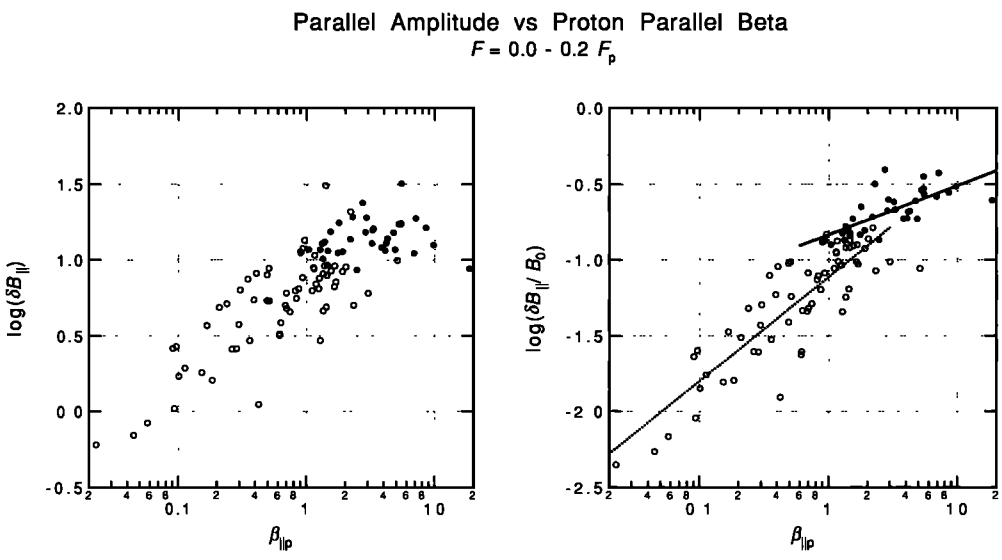

Figure 8. Parallel fluctuation amplitude versus $\beta_{\parallel p}$. Solid circles represent MIR and MRL events, for which the mirror mode is observed, whereas open circles represent the other events for which mirrorlike fluctuations were not prominent. The relative fluctuation amplitudes are ordered better, and the dependence of the MIR/MRL fluctuations on $\beta_{\parallel p}$ is different from the remainder of the data. See text for additional details.

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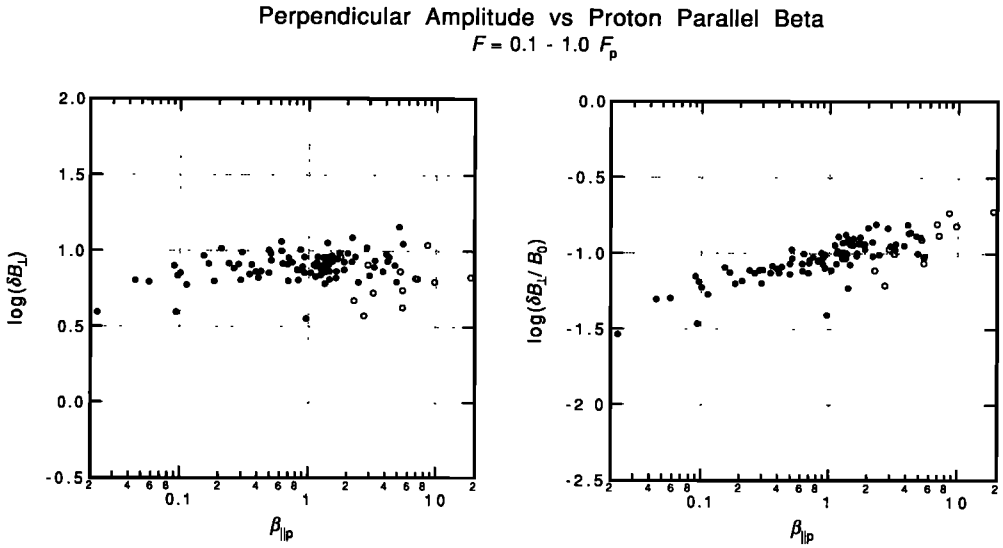

Figure 9. Transverse fluctuation amplitude versus $\beta_{\|p}$. Solid circles represent the BIF, CON, LOW, and MRL events for which ion cyclotron waves are present, and open circles show the MIR events during which cyclotron wave features were not observed. The relative amplitude exhibits a discernible trend for pulsation relative amplitude to increase with $\beta_{\|p}$.

We suggest that the evolution of spectral features may result naturally from the behavior of the mirror and cyclotron modes along the observed $A_p - \beta_{\|p}$ relationship. For $\beta_{\|p} << 1$ ($A_p >> 1$), linear theory predicts that the cyclotron instabilities have the lower anisotropy threshold [*Gary et al.*, 1993a] and that the proton and helium cyclotron modes grow in distinct frequency ranges, the proton mode growing for $F > 0.5F_p$ and the helium mode below $0.5F_p$ [*Denton et al.*, 1993]. This corresponds to the BIF spectra that occur for $\beta_{\|p} << 1$. For slightly higher $\beta_{\|p}$ (lower A_p), the frequencies of maximum growth rate of the proton and helium cyclotron instabilities decrease, but the proton mode frequency decreases more, leading to an overlap of the frequency regimes of the two cyclotron modes and the production of a single, continuous spectral peak. The plasma regime of overlapping proton and He^{2+} generated waves corresponds in $A_p - \beta_{\|p}$ space to the loci of the CON spectra. As $\beta_{\|p}$ (A_p) increases (decreases) further, the proton cyclotron mode frequency decreases below $0.5F_p$, leaving waves only below $0.5F_p$, and this regime is populated by LOW spectra. Since the He^{2+} and H^+ mode frequencies overlap, one cannot tell from the frequency structure which cyclotron mode is responsible for the single peak below $0.5F_p$. As $\beta_{\|p}$ (A_p) increases (decreases) yet further, the maximum growth rate of the proton cyclotron instability falls close to or below that of the mirror mode [*Gary et al.*, 1993a] and the two proton modes become competitive. Meanwhile, the He^{2+} cyclotron mode remains unstable. The simultaneous occurrence of mirror and cyclotron modes leads to the MRL spectra. Finally, for $\beta_{\|p} >> 1$ (A_p less than about 0.4), the mirror instability growth rate becomes larger than both cyclotron modes and dominates the spectral power giving MIR spectra.

For conditions that permit the cyclotron and mirror waves to coexist, computer simulations predict that the cyclotron modes play the leading role in governing the ion anisotropies despite their relatively small amplitudes [*McKean et*

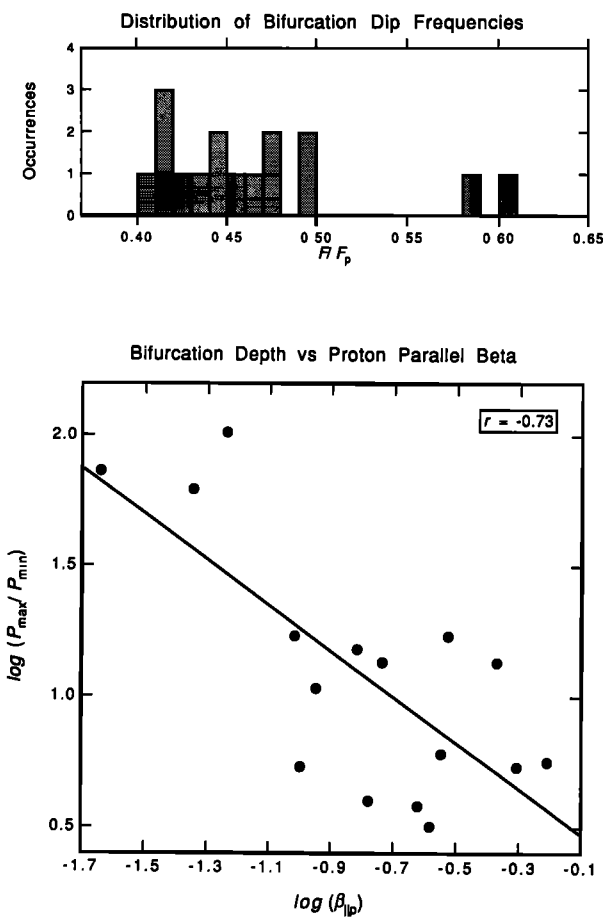

Figure 10. Frequency and depth of bifurcation feature. The upper panel shows the histogram distribution of bifurcation slot frequency. The slot appears between 0.4 and $0.5F_p$ as predicted by theory. The depth of the spectral slot versus $\beta_{\|p}$ is shown in the lower panel and shows that the slot becomes more pronounced for lower $\beta_{\|p}$.

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al., 1992; *Gary et al.*, 1993c]. The transverse amplitude increases with $\beta_{\|p}$ despite the fact that the transverse spectral features cease to be dominant, consistent with the prediction that the cyclotron modes are present even for high $\beta_{\|p}$.

In summary, we suggest that the general character of the fluctuations, compressive and transverse, the spectral structure and polarization of the transverse perturbations, and the correlation of spectral characteristics with location on the $A_p - \beta_{\|p}$ curve are all predicted theoretically [*Gary et al.*, 1993a; *Denton et al.*, 1993].

Implications for Order in the Magnetosheath

For the CCE data set, AF have shown that transverse fluctuations with $F > 0.5F_p$ and left-hand polarization correspond to the plasma depletion layer, whereas the predominantly compressional fluctuations correspond to the magnetosheath proper. The present analysis therefore implies that the progression of plasma and spectral properties corresponds to spatial structure in the magnetosheath. As plasma flows along the Earth–Sun line, $\beta_{\|p}$ and A_p change continuously from a high-$\beta_{\|p}$/low-A_p near the bow shock to a low-$\beta_{\|p}$/high-A_p close to the magnetopause.

The extent to which the magnetosheath plasma evolves toward the low-$\beta_{\|p}$/high-A_p extreme depends on the orientation of the magnetosheath field, lower $\beta_{\|p}$ occurring for northward field [AF]. It is natural to ascribe the difference to reconnection such that the additional flow diversion provided by reconnection relieves some of the burden for diverting the magnetosheath plasma.

Remaining Theoretical Challenges

The agreement between observed spectral structure and predicted frequency ranges of positive growth presented in Figure 1 is tantalizing. It is important to assess how well the linear Vlasov theory predicts the evolution of spectral structures more generally, perhaps using the $A_p - \beta_{\|p}$ relation. If linear theory continues to be successful, this will substantiate the mode identifications and the conclusion that the waves are generated locally. This theoretical study is under way.

Theoretical studies of the fluctuation amplitudes, polarization structure, and further relationships among the plasma parameters for comparison with the observations are of considerable interest but are more difficult. As *Denton et al.* [1993] have discussed, treatment of these issues requires that nonlinear saturation effects be included, which in turn implies that attention must be paid to the timescales of wave growth and residence with resonant particles. Hence quantitative predictions concerning spectral amplitudes and plasma responses to instability growth may come only from fully nonlinear computer simulations.

Progress addressing the nonlinear aspects of the problem has been made. The initial value simulations of *McKean et al.* [1992] show that the saturation value of mirror instability magnetic fluctuation energy increases with $\beta_{\|p}$; this shows the same trend as Figure 8. *McKean et al.* [1992] and *Gary and Winske* [1993] used simulations to study the saturation amplitude of the proton cyclotron instability. As $\beta_{\|p}$ is increased, with A_p adjusted so as to keep the linear growth rate constant, the saturation amplitude increases. This corresponds to Figure 9, which shows that the relative amplitude increases slightly with $\beta_{\|p}$.

The simulations of *McKean et al.* [1992] and *Gary and Winske* [1993] are initial value computations that require relatively large start-up anisotropies and therefore do not necessarily correspond to the observed magnetosheath conditions. The "refreshed" simulations of *Denton et al.* [1993] are probably a more appropriate way to address the model parameters, since the actual state of the plasma results from a dynamic equilibrium between plasma convection, which drives the anisotropy up, and the waves that reduce it. Preliminary results from the initial value computations provide reason for optimism that these aspects of the observations will also find direct explanation in terms of ion anisotropy instabilities.

6. Conclusions

We have characterized the magnetic spectral signatures and plasma properties observed in the subsolar magnetosheath for quasi-perpendicular upstream shock conditions. The magnetic spectral signatures and plasma properties were found to progress from mirror waves with high $\beta_{\|p}$ and low A_p in the magnetosheath proper to H^+ and He^{2+} cyclotron waves associated with high A_p and low $\beta_{\|p}$ in the plasma depletion layer.

Linear Vlasov theory has been used to test the interpretation that the mirror and cyclotron modes driven by the ion temperature anisotropies are responsible for the observed spectral features [*Gary et al.*, 1993a; *Denton et al.*, 1993]. Linear theory correctly predicts which modes should be observed and also appears to predict the spectral structure of the cyclotron waves. Furthermore, simulations of polarization and saturation amplitudes of the mirror and cyclotron modes indicate that these features also arise from the dynamics of these modes [*McKean et al.*, 1992; *Denton et al.*, 1993; *Gary and Winske*, 1993]. We conclude that the evolution of spectral structures may readily be explained by the generation of mirror and cyclotron instabilities in the plasma as it convects earthward from the shock to the magnetopause.

Additional evidence that the compressional fluctuations result from the mirror mode is provided by correlation analysis of density and field fluctuations. Density and field correlations show both that the field and density perturbations are in antiphase, and that the compressibility C_e is small, 0.13 on average, consistent with the expectation that $C_e < 1$ for mirror waves.

The widespread generation of ion anisotropy modes in the subsolar magnetosheath plasma implies that wave–particle interactions play a role in determining the ion anisotropy state of the plasma. This is clearly indicated by Figure 6. Since the ions carry the majority of plasma pressure in the magnetosheath, any process that regulates the distribution of ion pressures may be significant for magnetosheath dynamics. To accurately model the magnetosheath, proper account of the effects of the locally generated anisotropy instabilities needs to be made. We suggest that this include wave–particle effects using the relationship between A_p and $\beta_{\|p}$ as an the upper limit on the proton anisotropy.

Appendix A: Electron Density Fluctuations

Here we consider the relationship between fluctuations in the total electron density n_e and the density N_e observed

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by an instrument with a lower energy limit E_0. As discussed in the text, the electron temperature T_e responds adiabatically to changes in n_e, so that the effects of the fluctuating electron temperature need to be included. We therefore let the electron distribution function have the form

$$G(E) = \frac{n_e}{T_e} h(E/T_e) \tag{A1}$$

where

$$\int_0^\infty h(x)dx = 1$$

and temperature is expressed as an equivalent energy. The measured density is the integral of $G(E)$ above E_0, that is,

$$N_e = \int_{E_0}^\infty G(E)dE .$$

One can then write

$$\frac{\delta N_e}{N_e} = \frac{\delta n_e}{n_e} + \frac{\partial I(E_0,T_e)}{\partial T_e}\frac{\delta T_e}{I(E_0,T_e)} \tag{A2}$$

where

$$I(E_0,T_e) = \int_{E_0}^\infty h(E/T_e)\frac{dE}{T_e} = \int_{E_0/T_e}^\infty h(x)dx \tag{A3}$$

Using $n_e^{1-\gamma}T_e$ = const then gives

$$\frac{\delta N_e}{N_e} = \frac{\delta n_e}{n_e}\left[1 + (\gamma-1)\frac{E_0}{T_e}\frac{h(E_0/T_e)}{I(E_0/T_e)}\right] \tag{A4}$$

Recognizing that $h(E/T_e)$ is proportional to $E^{1/2}f(E)$, where $f(E)$ is the phase space density, we finally obtain

$$\frac{\delta N_e/N_e}{\delta n_e/n_e} = 1 + (\gamma-1)\cdot\frac{E_0^{3/2}f(E_0)}{\int_{E_0}^\infty E^{1/2}f(E)dE} \tag{A5}$$

which gives the desired relationship between $\delta N_e/N_e$ and $\delta n_e/n_e$ in terms of the measured electron distribution.

The special case of the Maxwellian distribution serves to illustrate the size of the effect. For a Maxwellian distribution,

$$h(E/T_e) = \frac{2}{\sqrt{\pi}}\left(\frac{E}{T_e}\right)^{1/2}\exp\left(-\frac{E}{T_e}\right) \tag{A6}$$

and A5 becomes

$$\frac{\delta N_e/N_e}{\delta n_e/n_e} = 1 + (\gamma-1)\cdot\frac{x_0}{1 - \frac{\sqrt{\pi}}{2}x_0^{-1/2}\exp(x_0)\mathrm{erfc}(x_0^{1/2})} \tag{A7}$$

where $x_0 \equiv E_0/T_e$ and erfc is the complementary error function. For $\gamma = 5/3$ and x_0 = 1, 2, and 3, this gives $(\delta N_e/N_e)/(\delta n_e/n_e)$ = 1.48, 2.10, and 2.74, respectively. Considering that the magnetosheath electron distributions are steeper than a Maxwellian in the vicinity of the thermal speed, the general expression of (A5) should yield somewhat larger values. Using (A5), we obtain 1.76 for the example and 2.25 on average, which is consistent with expectations for the HPCE electron instrument and magnetosheath electron temperatures in the 30- to 70-eV range.

Appendix B: Beta Definitions

To clarify notation, we offer the following guide to notation in our recent series of papers. The expressions here are in CGS units, and Boltzmann's constant is subsumed in the temperatures, that is, temperatures are in ergs. The MKS expressions used in the text of this paper and elsewhere use temperatures in kelvin and are obtained from the following relations by substituting $2\mu_0 k$ for 8π. Equation (A3) serves as a correction to the definition given by *Anderson and Fuselier* [1993, p. 1476].

In the work by *Gary et al.* [1993a,b]:

$$\beta_p = 8\pi n_e T_{\|p}/B^2 \tag{B1}$$

In the work by *Gary and Winske* [1993a,b] and *Gary et al.* [1993c]:

$$\beta_{\|p} = 8\pi n_e T_{\|p}/B^2 \tag{B2}$$

In the work by *Anderson et al.* [1991] and *Anderson and Fuselier* [1993]:

$$\beta = \beta_p = \frac{1}{3}8\pi n_p(2T_{\perp p} + T_{\|p})/B^2 \tag{B3}$$

In this paper,

$$\beta_p = \frac{1}{3}8\pi n_p(2T_{\perp p} + T_{\|p})/B^2 \tag{B4}$$

$$\beta_{\|p} = 8\pi n_p T_{\|p}/B^2 \tag{B5}$$

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